

From Model to Memory: Medium Alignment in Prosomoiosi (Simulation)

Liu Shuai

Academy of Media Arts Cologne (KHM)
Cologne, Germany
shuai.liu@khm.de

Mar Canet Sola

Baltic Film, Media and Arts School, Tallinn University
Tallinn, Estonia
mar.canet@tlu.ee

Abstract

Prosomoiosi (Simulation) is a real-time audiovisual exploration of how successive modelling frameworks overwrite cultural memory. Grounded in media archaeology, the work introduces the concept of *medium alignment*, arguing that ideas must be voiced through media whose operative logic remains visible. A live diffusion pipeline—combining TouchDesigner, StreamDiffusion, and ComfyUI Flux up-scaling—stages the algorithmic politics of selective remembrance and invites audiences to renegotiate authorship at the human-machine frontier.

CCS Concepts

• **Applied computing** → **Media arts**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

AI video, simulation, ComfyUI, media art, real-time generation

1 Introduction

Media and technology evolve in tandem, and every slight misalignment between them opens fertile ground for experimentation. Artificial intelligence can thus be regarded as a historical-technical agent that reshapes the conditions of aesthetic experience. Prosomoiosi (Simulation) situates this claim within contemporary image synthesis, demonstrating how real-time diffusion pipelines rewrite both the production and perception of moving images.

2 Concept

Simulation is not a simple act of copying but a power-laden rewriting produced through the mutual operations of technology, archives, and algorithms. From ancient sand tables to latent-space diffusion networks, modelling frameworks continuously install new regimes of seeing and forgetting. Building on this media-archaeological insight, the paper adopts the principle of *medium alignment*: concepts should be conveyed through media whose inner logic remains exposed rather than concealed. Two consequences follow. First, pushing AI image generation toward higher resolution and real-time

prompt editing reconfigures what qualifies as visual evidence. Second, examining database "selective memory" reveals how records are filtered and historicised, quietly shaping social structures and individual cognition.

3 Technical Description

3.1 Production Workflow

Found footage, gameplay recordings, and theoretical videos serve as generative foundations. Within TouchDesigner, the StreamDiffusion node (developed by dotsimulate) performs model-dependent transformations, yielding frame rates between 4 – 14 fps. Because this is below the standard 24 fps, source footage is first slowed and later re-accelerated so that generated and final frame counts match. The resulting sequence is exported frame-by-frame and processed in ComfyUI using the Flux up-scaling workflow.

3.2 ComfyUI Pipeline

The pipeline first enlarges each frame with the SIAx 4× super-resolution model. The upscaled image is then tiled via `TTP_Tile_image_size` (size estimation), `TTP_Image_Tile_Batch` (batch slicing), and `TTP_Image_Assy` (re-assembly). Each tile is refined by the Flux model with its LoRA in only four Euler sampling steps before the tiles are stitched back together, producing memory-friendly high-resolution results. Optionally, Topaz frame interpolation may be applied in post-production.

4 Conclusion

By aligning its conceptual questions with the technical affordances of real-time diffusion, Prosomoiosi demonstrates how medium alignment can foreground the infrastructural politics of simulation. The project argues that responsible artistic practice in the generative era must expose the mechanisms that steer perception while harnessing their creative potential.

Biographies

Liu Shuai is a Chinese digital media artist based in Cologne and Beijing. His practice spans algorithmic image-making, interactive games, and network installations, and his works have been screened at multiple international festivals.

Mar Canet Sola is a media artist and researcher at Tallinn University's Baltic Film, Media and Arts School, focusing on data-driven art and generative systems.

Acknowledgements

The authors thank Shuree Sarantuya and the Academy of Media Arts Cologne for technical support.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference'17, Washington, DC, USA

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnnn.nnnnnnn>

References

- [1] Jean Baudrillard. 1994. *Simulacra and Simulation*. University of Michigan Press.

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.